

Federated Explainable AI for Privacy-Preserving Cardiovascular Disease Detection Using Wearable Devices

Introduction

Artificial Intelligence (AI) continues to transform modern healthcare by enabling intelligent, connected, and patient-centered medical solutions. The integration of wearable technologies such as smartwatches, fitness trackers, and IoT-enabled medical devices has opened new possibilities for real-time health monitoring and early disease detection. However, traditional AI systems often rely on centralized data collection, creating concerns regarding privacy, transparency, and trust.

To address these challenges, Federated Explainable Artificial Intelligence (FedXAI) introduces a privacy-preserving healthcare framework that combines Federated Learning (FL) and Explainable AI (XAI). This approach allows medical data to remain securely on personal devices while enabling collaborative AI learning and interpretable decision-making.

The Need for Privacy-Preserving Healthcare

Cardiovascular diseases (CVDs) remain one of the leading causes of death globally. Continuous monitoring of heart conditions through wearable technologies offers opportunities for early diagnosis and timely intervention. However, transferring sensitive patient data to centralized cloud systems increases risks related to security breaches and regulatory compliance.

FedXAI solves this issue by allowing AI models to train directly on user devices. Instead of sharing raw health records, devices only send encrypted model updates to a central server, significantly improving privacy and maintaining patient ownership of medical information.

Core Architecture of FedXAI

The FedXAI framework operates through multiple intelligent stages:

1. Data Acquisition and Preprocessing

Wearable devices collect physiological signals including ECG, heart rate, blood oxygen levels, physical activity, and sleep patterns. These signals are cleaned, synchronized, and transformed into usable healthcare data.

2. On-Device Local Training

Each device independently trains an AI model using local patient information. This personalized learning improves prediction quality while preventing external access to sensitive records.

3. Federated Learning Aggregation

Encrypted model updates are securely transmitted and combined into a global learning model using federated aggregation techniques. Raw patient information never leaves the device.

4. Explainable AI Layer

Unlike traditional closed-box AI systems, FedXAI generates transparent explanations showing which health indicators influenced predictions. This improves trust among clinicians and patients.

5. Clinical Decision Support

The system produces interpretable health alerts and recommendations that support early diagnosis and proactive patient care.

Benefits of FedXAI in Healthcare

The framework delivers several important advantages:

- Strong patient data privacy protection
- Transparent and explainable AI decisions
- High diagnostic accuracy for cardiovascular monitoring
- Real-time health analysis using wearable devices
- Improved patient engagement and trust
- Scalable deployment across healthcare environments

Experimental findings indicate that FedXAI achieves diagnostic performance exceeding 90% across multiple healthcare sensor types while maintaining high transparency and privacy standards.

Future Opportunities

Future healthcare systems will increasingly rely on intelligent, decentralized, and explainable technologies. FedXAI demonstrates how AI can become more ethical, trustworthy, and clinically practical. Expanding this approach toward metabolic, neurological, and personalized healthcare applications may further transform global healthcare delivery.

Conclusion

Federated Explainable AI represents an important advancement in connected healthcare. By combining privacy-preserving learning with transparent AI decision-making, FedXAI enables secure and reliable cardiovascular disease detection using wearable devices. This innovation creates a future where patients maintain control over their health data while benefiting from advanced intelligent healthcare systems.